Slides for Patryk's Notes

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Outline

1. Naming conventions

2. TDL clarifications

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How we get the cumulant

$$\boldsymbol{G}^0(t)\boldsymbol{C}(t) = \iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0(t-t_1)\boldsymbol{\Sigma}^c(t_1-t_2)\boldsymbol{G}^0(t_2) \tag{1}$$

$$\implies C_{pp}(t) = i\int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega' t} \frac{\Sigma^c_{pp}(\omega'+\epsilon_p)}{(\omega'+i\eta)^2} \tag{2}$$

$$\implies C^{(2)}_{pp}(t) = \frac{1}{2}\sum_{iab} \frac{\langle pi \,||\, ab\rangle^2}{\left(\epsilon^{ab}_{pi}\right)^2}\left(e^{-i\epsilon^{ab}_{pi}t} + i\epsilon^{ab}_{pi}t - 1\right) + 2h1p \tag{3}$$

$$\implies A_{pp}(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\left(-i\Theta(t)e^{-i\epsilon_p t}e^{C^{(2)}_{pp}(t)}\right) \tag{4}$$

but Loos takes the first order cumulant, which presumably avoids numerical instability because it is just defined in the frequency domain

$$A_{pp}(\omega) = \frac{Z^{QP}_p}{\omega - \epsilon^{QP}_p + i\eta} + \sum_{iab}\frac{Z^{sat}_{piab}}{\omega - \epsilon^{sat}_{piab} + i\eta} + \sum_{ija}\frac{Z^{sat}_{pija}}{\omega - \epsilon^{sat}_{pija} + i\eta} \tag{5}$$

Naming conventions

- As eqn. 1 shows, the G_0 that defines both components in 4 should be the same.
- HF+C(2) makes sense and is probably clearer than HF+MP2+C
- HF+C(SRG-2) is not consistent, but SRG-MP2+C(SRG-2) is.
- We want to do BWs2+C(BWs2).
- This is why I prefer the notation $G_0@X; C(2@X)$.

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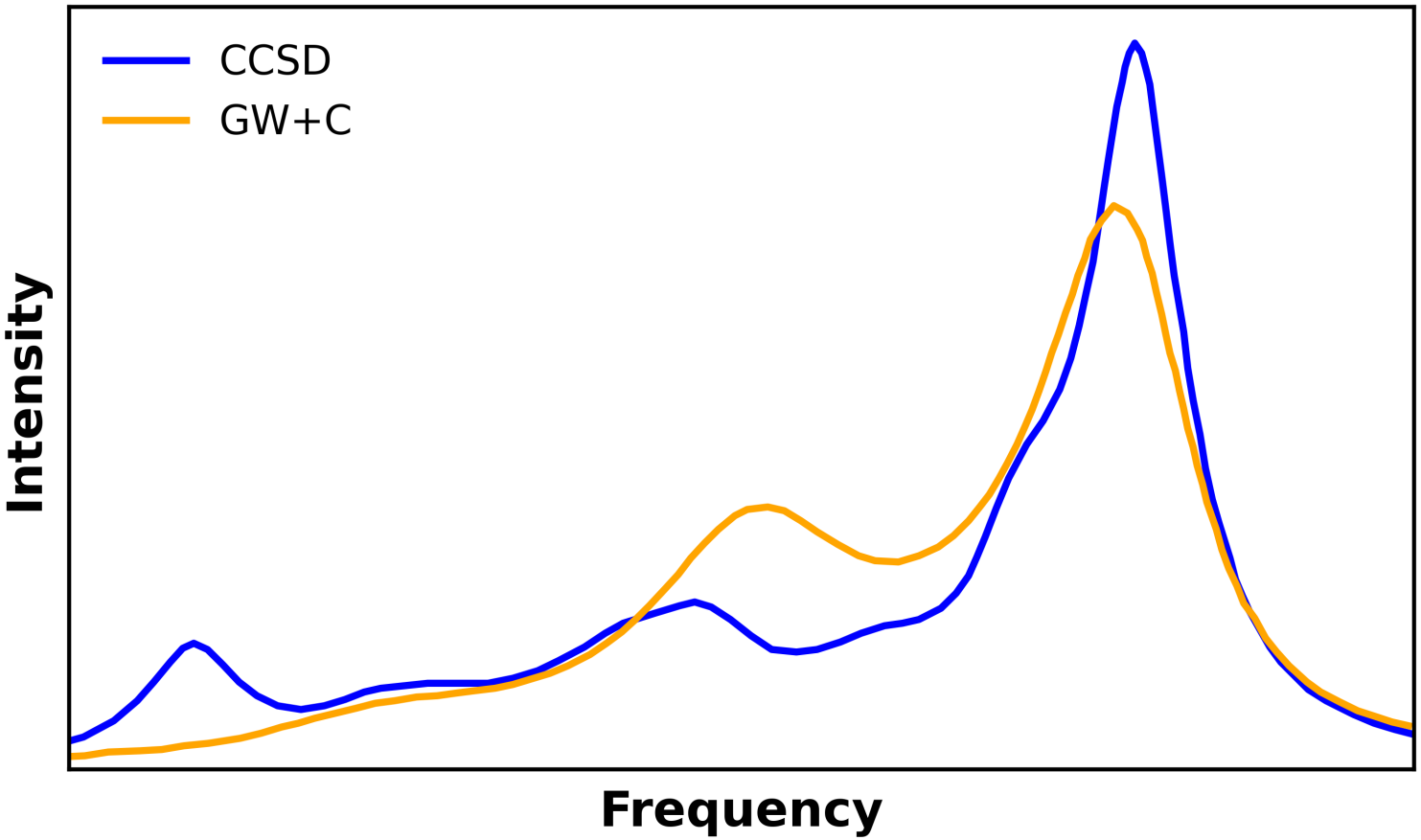
1. Naming conventions

2. TDL clarifications

What is Garnet's Fig. 2b showing?

- Why do you say we don't know the TDL spectral function at TDL?

Argument against the use of SRG-MP2 in extended systems

- Look at Kresse's argument
- Should I run SRG-MP2+C for rs=2 to check if this holds for cumulants?

$$G(t) = G^0(t)e^{C(t)}$$